Approach to Hematuria: Core Curriculum 2026

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Hematuria is a frequent clinical finding in primary care, nephrology, and urology clinics. Accurate diagnosis after a positive dipstick requires confirmation with urine microscopy, with the American Urological Association (AUA) guidelines defining significant microscopic hematuria as ≥3 red blood cells per high-power field. Hematuria warrants careful evaluation to distinguish between glomerular and nonglomerular etiologies, with attention to risk factors for malignancy to ensure timely diagnosis. Recent updates to the AUA guidelines emphasize risk stratification, advocating tailored diagnostic approaches to optimize detection of urologic cancers while limiting unnecessary procedures, health care costs, and patient harm. This review covers the evaluation and management of hematuria, including risk profiling for malignancy. Evidence-based recommendations are provided for common clinical scenarios such as patients with hematuria from malignancy, kidney stones, and glomerular disease. This core curriculum guides clinicians through the nuanced decision making required in the diagnosis and management of hematuria, with reference to the latest guidelines.

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Am J Kidney Dis. 88(4):605-614.

doi: 10.1053/j.ajkd.2026.05.013

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The Core Curriculum aims to give trainees in nephrology a strong knowledge base in core topics in the specialty by providing an overview of the topic and citing key references, including the foundational literature that led to current clinical approaches.

Introduction

Hematuria, the presence of red blood cells (RBCs) in the urine, is characterized as either gross hematuria that is visible to the naked eye or microscopic hematuria, which is detectable only with urine microscopy. Hematuria is common in primary care, nephrology, and urology clinics, making up about 1 in 5 urology evaluations.

When a patient reports hematuria, the first step is to confirm that the urinary system is the source of the blood. A urine dipstick includes an indicator pad impregnated with reagents that undergo a color change in the presence of heme or myoglobin. Thus, a positive dipstick can result from the presence of RBCs but also from free hemoglobin or myoglobin. Blood loss from other sources (eg, the gynecologic tract), the presence of systemic conditions such as rhabdomyolysis (myoglobinuria), and the ingestion of pigments from certain foods or substances (eg, rifampin or beets) can be mistaken for hematuria or even produce false positives on urine dipstick (eg, rhabdomyolysis). Vitamin C also can cause a false negative. For this reason, microscopic hematuria detected by dipstick requires a follow-up urine microscopy examination to confirm the diagnosis. After excluding the aforementioned conditions and confirming that the source of the RBCs is the urinary tract, the clinician should confirm the degree of hematuria. The American Urological Association (AUA) guidelines define microhematuria as ≥3 RBCs per high-power field on a single and properly collected urine specimen. If this definition is met, the hematuria is substantial enough to warrant additional work-up.

Once confirmed, hematuria demands an initial evaluation including a thorough history and physical examination with measurement of blood pressure and serum creatinine. During the history taking, one should consider the broad differential diagnosis of microscopic hematuria (Table 1) to focus the history on reviewing risk factors and signs of the potential etiologies. For example, one should inquire about symptoms of infection, a history of recent trauma or urologic instrumentation, and symptoms of kidney stones or prostate conditions. Genitourinary cancer is the most concerning cause of hematuria, so specific consideration of cancer risk factors such as age, biological sex, and extensive smoking history is mandatory (Box 1).

The clinician should also evaluate clinical factors that can help distinguish glomerular and nonglomerular etiologies for hematuria. For example, a family history of Alport syndrome or a recent upper respiratory infection, which can be seen with IgA nephropathy, can provide crucial clues to direct the work-up toward considering a glomerular cause. Urine microscopy can also provide important diagnostic clues. Findings of dysmorphic RBCs and RBC casts, the presence of additional abnormalities in the urine dipstick (ie, albuminuria), evidence of acute kidney injury, blood pressure elevations, and other systemic symptoms may indicate a glomerular cause. However, suspicion of glomerular bleeding should not negate a urologic work-up for most patients. Furthermore, given the smaller volume of blood with glomerular hematuria, the presence of blood clots is almost always

Table 1. Causes of Hematuria

Origin	Cause
Glomerular	
	Alport syndrome
	IgA nephropathy
	IgA vasculitis
	Thin basement membrane
	C3 glomerulopathy
	Lupus nephritis
	Antiglomerular basement membrane glomerulonephritis/disease
	Antineutrophil cytoplasmic antibody vasculitis
	Cryoglobulinemia
	Infectious glomerulonephritis
Nonglomerular	
Upper urinary tract	
Non-malignant	Nephrolithiasis
	Polycystic kidney disease
	Medullary sponge kidney
	Urothelial strictures
	Sickle cell trait or disease
	Pyelonephritis
	Renal infarct
	Renal tuberculosis
	Papillary necrosis
Malignant	Renal cell carcinoma
	Renal pelvis or ureteral transitional cell cancer
Lower urinary tract	
Non-malignant	Cystitis, prostatitis, urethritis
	Benign bladder or ureteral polyps
	Endometriosis of the urinary tract
	Urethral and meatal strictures
	Benign prostatic hyperplasia
Malignant	Prostate cancer
	Bladder cancer
Other	
	Heavy exercise
	Foreign body or chronic indwelling catheter
	Recent instrumentation or surgery
	Trauma or Injury to urinary tract
	Drug induced (eg, cyclosporine)

secondary to nonglomerular bleeding, so a urologic evaluation is warranted.

This installment of *AKJD*'s Core Curriculum in Nephrology covers the initial evaluation of patients with hematuria (Fig 1) and reviews the common causes of hematuria.

Additional Readings

➢ Barocas DA, Lotan Y, Matulewicz RS, et al. Updates to microhematuria: AUA/SUFU guideline (2025). *J Urol*. 2025;213(5):547-557. doi:10.1097/JU.0000000000004490 ★ESSENTIAL READING

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➢ Peterson LM, Reed HS. Hematuria. *Prim Care*. 2019;46(2):265-273. doi:10.1016/j.pop.2019.02.008

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Box 1. Common Risk Factors of Urothelial Cancer (List Is Not Exhaustive)

- Age
- Male sex
- Extensive smoking history
- History of gross hematuria
- Persistence of hematuria
- Degree of hematuria
- Prior pelvic floor radiation therapy
- Family history of urothelial cancer
- Family history of Lynch syndrome
- Occupational exposures to benzene chemical or aromatic amines
- Chronic indwelling foreign body in the urinary tract

*Based on information in Barocas et al, *J Urol*. 2025;213(5):547-557. doi:10.1097/JU.0000000000004490.

Risk Stratification

Case 1: *A 70-year-old man with hypertension, diabetes, and a 20 pack-year history of smoking presents to his primary care office for follow up after 2 episodes of gross hematuria in the past 3 months. He has no other symptoms. He has 50 RBCs per high-power field on a single office urinalysis. His serum creatinine is 1.0 mg/dL.*

Question 1: What is the risk stratification category for his overall risk of urothelial cancer?

(a) Low risk
(b) Intermediate risk
(c) High risk
(d) Indeterminate

Question 2: What are the next steps in a risk-appropriate management for this patient?

(a) Cystoscopy and abdominal ultrasound
(b) Cystoscopy and computed tomography urogram
(c) Urine cytology and computed tomography urogram
(d) Urine cytology and abdominal ultrasound

For the answers to these questions, see the following text

Malignancy is one of the greatest concerns for patients and providers when a patient presents with hematuria. In an observational study of 3,556 patients who were referred to urology with hematuria, bladder,

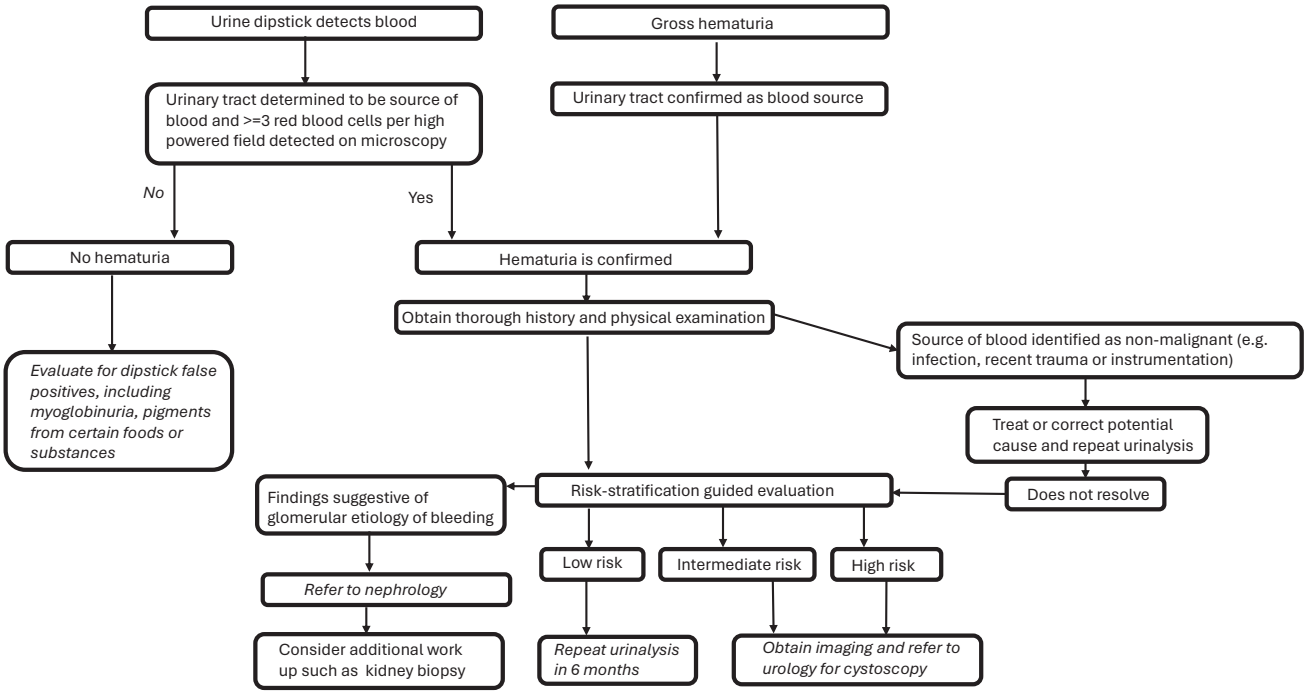


Figure 1. Evaluation of hematuria decision tree.

renal, or upper tract urothelial cancer was found in 13.2% of the patients with gross hematuria and 3.1% of patients with microscopic hematuria. Similarly, the aggregate urinary tract malignancy rate from 11 different studies dating from 2010 to 2019 was 1% among patients with microscopic hematuria and ranged from 0.3% to 6.25%.

To reduce the risk of missing a malignancy diagnosis, the older 2012 AUA guidelines recommended cystoscopy and computed tomography (CT) imaging for all patients over the age of 35 years who presented with microscopic hematuria. The updated 2025 guidelines use a nuanced approach that also considers the inherent risks of the more conservative approach. The risks for everyone of cystoscopy and CT include patient discomfort and infection during cystoscopy, radiation exposure from CT, iodinated contrast from CT urogram, risk of false-positive tests, and increased health care utilization and costs. This older man with a smoking history clearly fits into a high-risk category, so the answer to Question 1 is (c). The gold standard of imaging the upper tract with a CT urogram and then cystoscopy would be recommended, making (b) the answer to Question 2.

The 2025 guidelines focus on the importance of optimizing referrals so that patients at highest risk for malignancy undergo evaluation. Critically, current data suggest that opportunities for early detection are frequently missed. For example, one study noted that only 23% of patients with hematuria and risk factors for malignancy received appropriate imaging work-up and fewer (13%) underwent cystoscopy. Furthermore, certain patient populations are more likely to experience delay or not undergo a hematuria work-up compared with other populations (ie, women, African American patients).

The intention of the guideline was to reduce such delays and the underevaluation of patients at high risk for malignancy while limiting the risks and costs associated with overevaluation of patients at low risk. For this reason, the guidelines recommend using a risk stratification system to guide evaluation after the initial assessment. The risk stratification was developed from a systematic review of the urothelial cancer risk factor literature, and it was subsequently validated in 3 separate studies. We re-create the AUA Microhematuria Risk Stratification System from the AUA guidelines in Table 2.

Risk-Stratification Guides Work-Up

Risk stratification is a crucial first step because this will guide the appropriate work-up. The guidelines recommend that low-risk patients be monitored with a repeat urinalysis in 6 months. If hematuria remains persistent then patients are reclassified to a higher level of risk. Intermediate-risk patients are recommended for cystoscopy and kidney ultrasound, and high-risk patients are recommended for cystoscopy and upper urinary tract axial imaging with delayed phase imaging such as CT or magnetic resonance imaging.

Urinary Markers

Urinary markers (cytology or urine-based tumor markers) should not be routinely used to make decisions about whether to perform cystoscopy due to the variable

Table 2. Risk Stratification for Microhematuria

Criteria	Low Risk	Intermediate Risk	High Risk
Degree of hematuria, red blood cell/high-power field	3-10	11-25; previously low risk with no prior evaluation and 3-25 on repeat urinalysis	>25; history of gross hematuria
Age, years	<60 years old, women; <40 years old, men	≥60 years old, women; 40-59 years old, men	≥60 years old, men
Smoking history	Never smoker, <10 pack-year	10-30 pack-years	>30 pack-years
Presence of additional urothelial cancer risk factors[a]	None	Any	One or more

Patients must meet all criteria in the low-risk group but only need to meet 1 criterion in the intermediate and high-risk groups. Age for women should not be used as a sole criterion for placing patients at high risk.
Reproduced in modified form from Barocas et al, *J Urol*. 2025;213(5):547-557. doi:10.1097/JU.0000000000004490 with permission of the copyright holder (Wolters Kluwer Health).
[a]Urothelial cancer risk factors included in the risk stratification system include: age, male sex, smoking use, degree and persistence of microhematuria, history of gross hematuria.

sensitivity and specificity of these tests. For example, a meta-analysis that included 1,255 patients and 18 studies showed that using cytology to detect urothelial cancer had an overall sensitivity of 34%. Other meta-analyses have found that other tumor-based markers perform better than cytology though they failed to perform better than cystoscopy. Furthermore, the data do not support the addition of urinary markers in the setting of negative cystoscopy. For example, in a study of 2,778 patients with microscopic hematuria, only 2 patients who had a negative cystoscopy and imaging but had positive cytology were ultimately diagnosed with urothelial carcinoma.

Bladder Cancer

Bladder cancer, the 10th most common cancer worldwide and in the United States, is predominantly urothelial carcinoma. Bladder cancer is more common in men over the age of 65 years. The most common risk factors include smoking and occupational risk factors such as aromatic amines and polycyclic aromatic hydrocarbons. Any prior smoking has a population attributable risk of 0.50 in men and 0.52 in women. Smoking cessation can reduce the risk of bladder cancer, but their risk remains higher than for nonsmokers; secondhand smoke exposure also increases risk. Other less common but described risk factors include human papilloma virus infection, recurrent urinary infections, *Schistosoma haematobium* infection, chronic inflammation (chronic indwelling catheters), cyclophosphamide exposure, and type 2 diabetes.

As with other cancers, genetic factors can directly and indirectly affect bladder cancer risk. Prior genomewide association studies used 24 independent markers to developed a polygenic risk score that is associated with increased risk of cancer for individuals independent of smoking history. Additionally, there are some data showing that family history increases the individual risk for bladder cancer.

The most common presenting symptom of bladder cancer is painless intermittent gross hematuria. However, in locally advanced or metastatic disease, patients can be symptomatic with irritative or obstructive voiding symptoms, pain, and constitutional symptoms. A diagnosis of bladder cancer typically involves an office-based cystoscopy. Urothelial abnormalities identified on cystoscopy require either an endoscopic biopsy or a transurethral resection of the abnormality for a pathologic diagnosis. Tumor size along with grade and depth of tumor invasion provide important prognostic and staging information and guide treatment planning.

For noninvasive disease (not invading the muscle), patients may then be considered for intravesical chemotherapy or immunotherapy with bacillus Calmette-Guerin. For invasive disease, patients may then be considered for either radical cystectomy with neoadjuvant or adjuvant therapy (with cisplatin-based chemotherapy) or trimodal therapy with maximal endoscopic resection followed by concurrent chemotherapy and radiotherapy.

Hematuria Follow-Up

Patients who have a negative malignancy that is a risk-stratified based evaluation and have negative urinalysis in a follow-up evaluation do not need further urologic evaluation. Many clinicians find it challenging to determine the need for repeat evaluation in patients who have persistent hematuria but a negative initial urologic evaluation. The majority will not have an etiology identified in follow-up evaluations. A recent retrospective study found that of 637 patients with microscopic hematuria and a negative work-up (161 [25%] of whom had repeat cystoscopy and 317 [50%] of whom had repeat upper tract imaging at median 39 months), 2 (1.2%) had a new bladder malignancy identified and 4 (1.3%) had a new renal mass identified.

The AUA guidelines recommend a shared decision making strategy that is based on the individual's malignancy risk. Many providers will repeat a urinalysis annually for high-risk patients and every 5 years for low-risk patients. If patients with microhematuria and a prior negative

evaluation then develop gross hematuria or an increase in microhematuria, patients should have a repeat complete evaluation with cystoscopy and imaging.

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Diagnosis of Kidney Stones

Case 2: *A 35-year-old man presents with a 2-day history of right flank pain radiating to the groin. He has no fever, but he is experiencing nausea and vomiting. CT imaging reveals a 6-mm stone at the right ureterovesical junction without evidence of hydronephrosis. About 6 months ago, he had 2 episodes of gross hematuria with flank pain after which he spontaneously passed a small kidney stone. He retrieved and saved the prior stone.*

Question 3: What is the most appropriate strategy for acute management?

(a) Observation and pain control
(b) Medical expulsive therapy and pain control
(c) Urgent ureteroscopy
(d) Urgent percutaneous nephrolithotomy

Question 4: In addition to a basic metabolic panel with serum calcium, which of the following is appropriate for the initial stone risk evaluation for this patient when he presents for post–emergency department follow up in the clinic?

(a) 24-hour urine collection
(b) Urinalysis
(c) Urinalysis, 24-hour urine collections, and parathyroid hormone
(d) Urinalysis, 24-hour urine collections, and a stone composition analysis

For the answers to these questions, see the following text.

Kidney stones are often diagnosed or confirmed by imaging that is obtained in the setting of hematuria. Other clinical indications of kidney stones include flank pain or a previous history of kidney stones. Imaging modality choices to evaluate kidney stones include CT, ultrasound, and X-ray (kidney, ureter and bladder: KUB), and each has variable sensitivity and specificity. Noncontrast CT is the study of choice for imaging as part of the initial evaluation of patients who have acute onset of flank pain or suspicion of kidney stones. This is because the sensitivity and specificity of CT is much higher than other imaging modalities. Sensitivity of CT is reported to be as high as 0.95 and specificity as 0.97. However, in routine monitoring of stone burden, it is common to obtain ultrasound and KUB in order to reduce radiation exposure.

Kidney Stone Epidemiology

Kidney stones are common, and their prevalence has been rising—over 10% of individuals in the United States will get a kidney stone in their lifetime. Of patients presenting with hematuria, one retrospective review found that over 11% have a kidney stone as the etiology of their microscopic hematuria. The classic presentation of a kidney stone is flank pain and hematuria. The majority, but not

all, of patients with acute presentation of a kidney stone will have hematuria. For example, one study found that about 68% of patients with a ureteral stone will have subjective hematuria. Additionally, the timing of pain onset and the urine collection may affect the sensitivity of hematuria. In a study of kidney stone patients with CT-confirmed stones, the sensitivity of hematuria changed over time such that it was highest (95%) on day 1 and lowest at days 3 and 4 (65% and 68%, respectively) from pain onset. Notably, however, in a large retrospective study comparing patients presenting with renal colic both with and without hematuria, there was no difference in time to presentation based on the presence of hematuria.

Risk Factors and Pathophysiology of Kidney Stones

There are many important clinical risk factors for kidney stones. These risk factors are commonly related to underlying genetics, systemic medical conditions, medications, and/or environmental or dietary exposures. For example, family history, diabetes, certain bowel diseases, and a low-calcium diet are all strongly associated with a higher risk of kidney stones. These clinical risk factors change the urine composition such that they increase the supersaturation of the urine with respect to the stone material (ie, calcium oxalate, calcium phosphate, or uric acid). By increasing the supersaturation, these factors then increase the risk of new stone formation or growth in the urinary space. A low-calcium diet, for example, increases urinary oxalate excretion and subsequently the calcium oxalate supersaturation and kidney stone risk.

Once a kidney stone forms it can cause both gross and microscopic hematuria. The hematuria from kidney stones likely results from damage to the mucosa of the upper or lower urinary tract from mechanical irritation, inflammation, and/or blockage.

Acute Management of Kidney Stones

The first consideration for acute management of a kidney stone passage event is assessing urgency of a urological consultation. The presence of signs including urinary tract infection (UTI), fever, acute kidney injury, uncontrollable pain, or obstruction on imaging requires an urgent urologic consultation and consideration for urgent urologic intervention.

The majority of patients will not have these symptoms and can be managed conservatively in the outpatient setting. Patients may benefit from referral to urology for nonurgent urologic management in the outpatient setting. The 2 principles of conservative management are pain control and monitoring for stone passage.

Nonsteroidal anti-inflammatory agents (NSAIDs) remain the foundation of management of kidney stone–related pain. A 2025 Cochrane systematic review that included 3,593 patients from 29 different trials found that NSAIDs were effective at reducing pain compared with placebo. Additionally, in a 2018 systematic review that included 4,887 patients and 36 trials, NSAIDs were slightly superior to opioids in pain reduction at 30 minutes and had an improved side-effect profile with less vomiting compared with patients treated with opioids. However, NSAID use should be limited in patients with reduced estimated glomerular filtration rate at baseline and are not recommended for patients with acute kidney injury.

The likelihood of stone passage without a urologic intervention is heavily dependent on stone size. Stones under 5 mm are likely to pass on their own without pharmacologic therapy. Stones greater than 5 mm may require intervention to pass about 50% of the time. For stones between 5 mm and 1 cm, the likelihood of requiring intervention increases, and medical expulsive therapy (MET) is commonly prescribed. Because the patient in this case has a 6-mm stone, MET and pain control is an appropriate acute management strategy, so the answer to Question 3 is (b).

MET typically includes α-blocker medications, such as tamsulosin. Although the usefulness of MET has been debated, meta-analyses and a Cochrane review have shown that 4 weeks of MET with an α-blocker may help spontaneous passage, and the AUA guidelines support the use of MET with α-blockers for uncomplicated ureteral stones ≤10 mm. Stones larger than 1 cm are unlikely to pass spontaneously or with MET and will likely require urologic intervention. For all stone sizes, the time to stone passage can vary, with some stones taking several weeks to pass.

When following a patient for spontaneous passage or on MET, it is important to consider imaging to confirm that the patient has passed the stone. Some patients will have asymptomatic obstruction, and the clinical symptoms (eg, cessation of pain) may not accurately predict the resolution of hydronephrosis or stone passage.

Surgical intervention is warranted for patients who have prolonged obstruction (4-6 weeks), have large volume stone disease, or are unable to tolerate conservative management (ongoing pain, nausea, vomiting, kidney dysfunction). The common surgical intervention options for stone disease include shockwave lithotripsy, ureteroscopy with laser lithotripsy, and percutaneous nephrolithotomy.

The decision on the specific intervention modality is based on stone factors such as the stone size, type, location, and density; patient factors such as kidney function, kidney anatomy, and comorbidities; and the goals of treatment. In general, ureteroscopy and shockwave lithotripsy are effective treatments for smaller stones whereas a percutaneous approach is reserved for larger volume stones (>2 cm) or stones that fail treatment with less invasive approaches. In recent years, ureteroscopy has become the most commonly performed surgical intervention for stone disease. This is likely due to the versatility of ureteroscopy, advances in surgical training, and improvement in ureteroscope and laser technology.

Preventative Strategies for Kidney Stones

Kidney stone prevention is also a crucial part of management for patients who form kidney stones. Referral to nephrology for a metabolic work-up is recommended for recurrent kidney stone patients and for patients who have a solitary kidney; and it also should be considered for other groups who are at high risk for recurrence, such as those with cystinuria or uric acid stones.

The initial metabolic work-up for recurrent kidney stone patients includes a basic metabolic panel with serum calcium, urinalysis, 24-hour urine collections, and a stone composition analysis (if available), so the answer to Question 4 is (d). Given the day-to-day variation in dietary patterns, it is common to obtain two 24-hour urine collections during the initial work-up. The 24-hour urine collections are critical for preventative therapy strategies because they provide clinical data to guide targeted prevention strategies based on the individual's specific urinary risk factors, such as hypercalciuria, hyperoxaluria, or hypocitraturia. Additional work-up, including parathyroid hormone levels and genetic testing is recommended if indicated after initial testing.

Individualized Strategies for Kidney Stone Prevention

A prevention plan that is individualized to a patient's 24-hour urine collection is associated with improved kidney stone outcomes with lower rates of recurrence and procedures. The overall goal is to reduce the urine supersaturation of the stone type (eg, calcium oxalate, calcium phosphate, or uric acid) as this is expected to reduce the risk of stone formation. To do so, one should identify which modifiable parameters in the 24-hour urine are contributing to a high supersaturation. For example, if urine calcium levels are high, then one should consider high sodium or protein intake as potential contributors to the high urine calcium by examining the 24-hour urine sodium and protein catabolic rate. The dietary advice should prioritize the factor(s) contributing to high urine calcium for that individual. This may include reducing dietary sodium (if the sodium intake is greater than 2,300 mg/day) or reducing protein intake (if the protein intake is above moderate). Once dietary modifications are optimized, pharmacologic therapies should also be considered.

After initiating a preventative strategy, a single repeat 24-hour urine collection can evaluate the success of the strategy and guide the ongoing preventative strategy. A similar approach is recommended for other urinary abnormalities associated with increased supersaturation and kidney stone risk, such as high urine oxalate, low urine citrate, and low urine pH.

Additional Readings

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Differential Diagnosis of Glomerular Causes of Hematuria

Case 3: *A 23-year-old woman presents for evaluation of recurrent microscopic hematuria. Initially the hematuria was identified in the setting of a UTI, but she has continued to demonstrate >20 RBCs per high-power field on evaluation after the resolution of the UTI. Her kidney function is normal, and there is no evidence of proteinuria. Her family history includes a twin brother with hearing loss, and her mother has been told that she has blood in her urine, which was never evaluated further.*

Question 5: What is the most appropriate next step in the diagnosis of this patient?

(a) Immunoglobulin A levels
(b) Genetics panel
(c) No additional work-up
(d) Urine culture

For the answer to this question, see the following text.

In glomerular-related hematuria, immune-mediated injuries to the glomerular capillary endothelium, non-immune-related structural abnormalities in the glomerular basement membrane, or gaps in the capillary walls allow RBCs to cross the basement membrane and enter

Bowman's space. In adults, the most common glomerular causes of hematuria are immunoglobulin A (IgA) nephropathy, Alport syndrome, or thin basement membrane disease. The full differential diagnosis will also include less common conditions such as lupus nephritis and C3 glomerulopathy.

Diagnosis of Glomerular Hematuria

Glomerular etiologies of hematuria are more likely when dysmorphic RBCs or RBC casts, proteinuria, or impairment in kidney function are also present. If a patient with glomerular bleeding has gross hematuria, a brown, cola-colored urine may be present. In this setting, a kidney biopsy may be indicated.

During an initial work-up for hematuria, as a part of the history and physical examination one should screen for specific features that are suggestive of glomerular causes of the hematuria. For example, a family history of hematuria could be suggestive of thin basement membrane or Alport syndrome. A family history of kidney failure of unclear etiology may be suggestive of Alport syndrome. A personal or family history of sensorineural hearing loss may be consistent with Alport syndrome. A history of acute upper respiratory infection preceding onset of hematuria may be consistent with IgA nephropathy.

The initial serologic work-up should include an estimate of kidney function. Complement levels and the antinuclear antibody should be considered to complete the evaluation for the less common glomerular etiologies of hematuria, C3 glomerulopathy, and lupus nephritis. Additionally, depending on the clinical situation, one can consider obtaining double-stranded DNA, antineutrophil cytoplasmic antibodies, and antiglomerular basement membrane antibody.

To establish a diagnosis, a kidney biopsy may be indicated. Clinicians should participate in shared decision making with the patient about the risks and benefits of a kidney biopsy, but the biopsy may be indicated in settings of positive serologies, proteinuria, hypertension, reduced kidney function, or during evaluation for living donor transplantation. Persistent unexplained microscopic hematuria may also be an indication for kidney biopsy if a definitive diagnosis is desired, once urologic causes have been excluded. Finally, genetic testing may be an important part of the diagnosis for conditions such as Alport syndrome.

IgA Nephropathy

IgA nephropathy is the most common cause of primary glomerular disease worldwide, and it is caused by deposits of IgA in the glomerular mesangium. A study in the People's Republic of China found that among 13,519 kidney biopsies, over 45% were IgA nephropathy. A separate study of biopsy diagnoses found that over 22% of diagnoses from Europe and 11% from North America were IgA nephropathy.

The clinical features of IgA nephropathy can vary because there is a large degree of clinical and pathological heterogeneity. Children and young adults in North America commonly present with macroscopic hematuria during another illness (ie, respiratory or gastrointestinal) whereas adults typically present with microscopic hematuria, with or without proteinuria, hypertension, or reduced kidney function. IgA nephropathy is also commonly associated with other conditions, such as chronic liver disease, autoimmune conditions, and viral infections (eg, hepatitis or HIV).

The diagnosis of IgA nephropathy requires a kidney biopsy because there are no validated serum or urine biomarkers for IgA nephropathy. The Kidney Disease Improving Global Outcomes (KDIGO) 2025 Clinical Practice Guidelines on the management of IgA nephropathy recommend considering kidney biopsy for patients where IgA nephropathy is possible and urine protein is at least 0.5 g/day. Further, once the diagnosis of IgA is established, it is recommended to screen for secondary causes such as liver disease.

Alport Syndrome

Alport syndrome is a genetic disease caused by variants in α-chains of type IV collagen of the basement membranes of the kidney, eye, and cochlea. In the United States, the incidence of Alport syndrome is estimated at about 3% of children and 0.2% of adults with end-stage kidney disease. Genetic transmission of Alport syndrome is most often X-linked with variants of the *COL4A5* gene, but it can also be autosomal recessive or autosomal dominant with variants of either the *COL4A3* or *COL4A4* genes. Additionally, disease penetration varies based on environmental and other genetic factors.

The clinical features of Alport syndrome reflect the distribution of affected type IV collagen in the basement membranes in the kidney, eye, and cochlea. These gene variants lead to heterogeneous abnormalities, including glomerular disease (microscopic hematuria, progressive kidney disease, and end-stage kidney disease), ocular abnormalities, and sensorineural deafness.

Alport syndrome commonly presents with persistent asymptomatic microscopic hematuria in childhood. Men presenting with classic X-linked disease will have more advanced disease while patients presenting with autosomal dominant disease or women with X-linked disease may have a variable clinical presentation and course.

For patients who have a positive family history of Alport syndrome or who have physical examination features consistent with Alport syndrome (ie, sensorineural hearing loss or ocular findings), the molecular genetic testing can confirm the diagnosis. For this woman with microscopic hematuria and a family history of hearing loss, a genetics panel can confirm the diagnosis, so the correct answer is (b). Biopsy of skin or kidney tissue can also lead to diagnosis. This information can also be used for genetic

counseling and prognostication. The molecular genetic testing of *COL4A3*, *COL4A4*, and *COL4A5* has good sensitivity and specificity.

Thin Basement Membrane Nephropathy

Thin basement membrane nephropathy (TBMN) is a common condition that has a strong genetic component and is characterized by diffuse thinning of the glomerular basement membrane. It has some similarity to Alport syndrome: heterogenous variants to *COL4A3* and *COL4A4* have been identified in some, but not all, patients and families with TBMN. Although it was previously considered a benign condition, it has been associated with focal segmental glomerulosclerosis.

The main clinical feature of TBMN is asymptomatic microscopic hematuria. Patients typically present without other abnormalities though some may present with microscopic hematuria and mild to moderate proteinuria or hypertension. A kidney biopsy with electron microscopy will confirm the diagnosis, but not all patients require a biopsy. The diagnosis can often be based on the benign clinical presentation (ie, isolated hematuria without proteinuria or reduced kidney function) in the setting of a family history of hematuria (without kidney failure). If a patient with a TBMN diagnosis develops new or worsening clinical features, such as worsening proteinuria, it is important to consider other conditions such as Alport syndrome or IgA nephropathy. It is possible that Alport syndrome may be initially diagnosed as TBMN if the biopsy is performed early.

Treatment

The treatment for glomerular causes of hematuria depends on the final diagnosis.

For IgA nephropathy, the KDIGO guideline for IgA nephropathy management was recently updated. It is recommended to assess the risk of progression using histologic findings from the biopsy, the estimated glomerular filtration rate, and the degree of proteinuria. Based on the risk of progression and overall treatment goals, the treatment strategies include general supportive care with or without immunosuppressive therapy. General supportive care includes renin-angiotensin system (RAS) blockade, blood pressure control, lifestyle modifications (eg, low sodium diet, smoking cessation), and consideration of additional therapies to target both proteinuria and the immune pathogenesis of the disease.

For Alport syndrome, clinical practice guidelines were also recently updated. In this update, RAS blockade remains the standard of care. RAS blockade should be initiated at diagnosis for both men with X-linked and for individuals with autosomal recessive Alport syndrome. Women with X-linked and all individuals with autosomal dominant Alport syndrome are recommended to have RAS blockade initiated at the onset of albuminuria. RAS blockade is standard because studies have found that RAS blockade delays end-stage kidney disease and increases the patient's life expectancy compared with no treatment. Other therapies, such as cyclosporine, have shown inconsistent data and are not recommended.

Patients with Alport syndrome who undergo kidney transplantation have excellent outcomes and do not have disease recurrence. However, it is recommended to follow them closely, particularly in the first year after transplant, and to have a low threshold for allograft biopsy because about 3% of men can acquire posttransplant anti-glomerular basement membrane disease.

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Sickle Cell Trait

Case 4: *A 25-year-old man presents to clinic after a self-limited episode of painless, gross hematuria. This has occurred before, including 1 severe presentation where he urinated dark blood clots and experienced weakness and fatigue. Each prior episode resolved without intervention. He has a brother with sickle cell disease.*

Question 6: What is the most likely cause of this patient's hematuria?

(a) Non-urothelial bladder cancer
(b) Papillary necrosis
(c) Renal medullary carcinoma
(d) Kidney stone

For the answer to this question, see the following text.

Epidemiology

Individuals with sickle cell trait are at risk of urologic and kidney disease including hematuria, chronic kidney disease, albuminuria, and renal medullary carcinoma. A

systematic review found that episodes of hematuria are common in patients with sickle cell trait (2.5% vs 1.3%), a nearly 2-fold increase in the relative risk, compared with those without sickle cell trait; however, the strength of the evidence was low because it was based on a single cohort of moderate quality.

Pathogenesis of Hematuria in Sickle Cell Trait

Patients with sickle cell trait may develop hematuria from multiple causes. Papillary necrosis or infarct is a common cause of hematuria and is the most likely cause of this patient's hematuria, so the answer to Question 6 is (b). However, the differential diagnosis includes UTIs and renal medullary carcinoma. Patients with sickle cell trait, particularly if pregnant, may be at higher risk for UTIs and pyelonephritis with an adjusted odds ratio of 1.98 (95% CI, 1.02-3.85) compared with women without sickle cell trait. Finally, although the condition is rare, patients with sickle cell trait and hematuria also need to be evaluated for renal medullary carcinoma, particularly when there is no other clear explanation for the hematuria. Renal medullary carcinoma is an aggressive non-clear cell renal cancer that is associated with sickle cell trait and continues to have poor survival outcomes, despite new therapies.

Diagnosis

When evaluating a patient with sickle cell trait and hematuria, a urinalysis with microscopy and culture should be included in the diagnostic work-up. Given the high risk of urothelial malignancy due to the risk of renal medullary carcinoma, these individuals should also be considered for upper urinary tract imaging. There are no data to guide the optimal choice of imaging, but most experts recommend obtaining CT imaging of the abdomen and pelvis.

Treatment

Management of hematuria from a UTI is similar to management of a UTI in a patient who does not have sickle cell trait. Management of papillary necrosis is supportive. For the rare patient with renal medullary carcinoma, the current standard of care therapies include nephrectomy and platinum-based chemotherapy.

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Conclusion

Hematuria is a common and clinically significant finding across primary care, nephrology, and urology practices. Prompt and accurate diagnosis is essential to distinguish between benign and serious underlying causes, including malignancy, glomerular disease, nephrolithiasis, and more rare genetic syndromes. The current AUA guidelines advocate a risk-stratified approach to evaluation of hematuria, ensuring that high-risk patients receive comprehensive diagnostic evaluation in a timely manner while minimizing unnecessary procedures in low-risk individuals.

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Support: None.

Financial Disclosure: The authors declare that they have no relevant financial interests.

Peer Review: Received October 29, 2025, in response to an invitation from the journal. Evaluated by 4 external peer reviewers and a member of the Feature Advisory Board, with direct editorial input from the Feature Editor and a Deputy Editor. Accepted in revised form May 17, 2026.